

K-FIELD EARTH KINEMATIC DERIVATION OF THE COUPLING CONSTANT

Earth-only calibration from observed eccentricity-vector motion and apsidal evolution under
the origin-locked ICRS Cell3 force law

Principal Investigator: P. Dwayne Esterline

BS, Huntington University (1999) | MBA, Indiana Wesleyan University (2008)

Krytronx

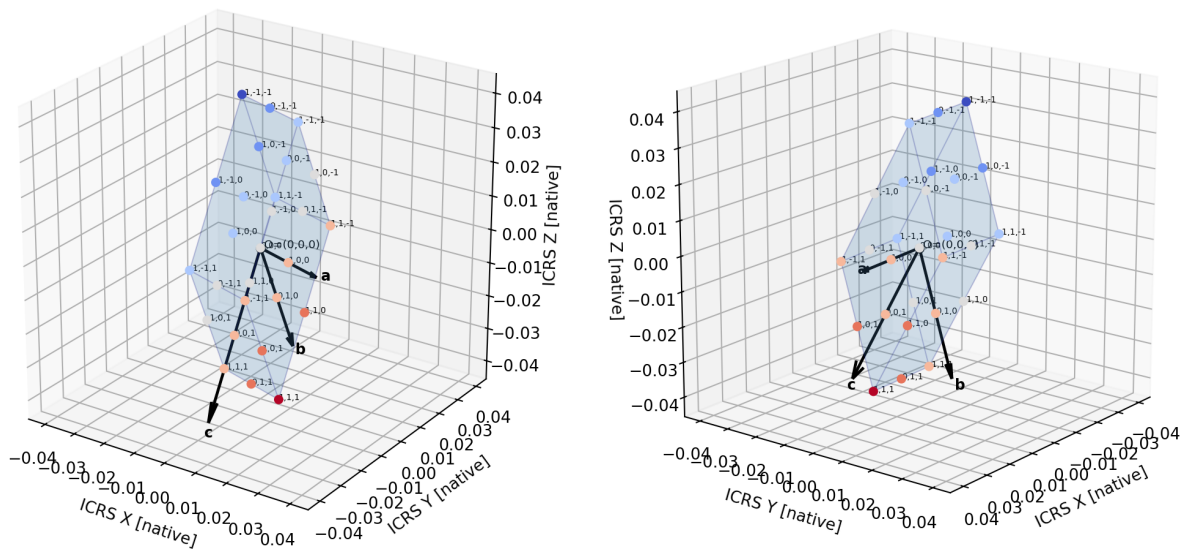
Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.

Generated: Unix epoch 1784828045 | UTC 2026-07-23T17:34:07Z

Origin-locked Cell3 base cell: complete node and boundary geometry

View 1: all 27 half-step nodes

View 2: all 27 half-step nodes



Title figure. The complete origin-locked ICRS Cell3 base cell, including all 27 centered half-step nodes, six boundary planes, and the primitive basis vectors. The figure is generated directly from the machine-readable geometry used in the Earth coupling derivation.

Abstract

This reference derives the K-Field coupling constant from Earth kinematics alone. The Cell3 translational gauge is fixed at the ICRS Cartesian origin $O=(0,0,0)$, while the locked primitive orientation and phase metric are retained. The Earth-Sun differential K acceleration is computed daily for a complete 366-epoch year, resolved in radial, tangential, and normal orbital coordinates, and integrated into the Earth eccentricity-vector response. The observed Earth-Moon-barycenter eccentricity magnitude, inclination, longitude of perihelion, and their fitted rates are transformed from the J2000 mean ecliptic frame into ICRS vector form. The resulting observed one-year eccentricity-vector change is projected onto the calculated unit-coupling K response. This full-vector calibration gives $\kappa_F=3.0428e-11$. An independent apsidal-rate calibration gives $\kappa_F=3.1054e-11$. Their 2.1% agreement establishes the Earth-only required interval $3.0e-11$ to $3.1e-11$, with recommended central value $\kappa_F=3.0741e-11$. At this coupling, the Earth-Sun differential K acceleration spans $5.68e-10$ to $7.08e-10$ m/s² and corresponds to a force scale of $3.39e15$ to $4.23e15$ N for Earth mass. The positive sign is the branch that reproduces the observed K-indexed apsidal and eccentricity-vector direction.

Quantity	Locked or derived value
Coordinate frame	ICRS/J2000 Cartesian
Cell3 origin	$O=(0,0,0)$ ICRS
Observed Earth eccentricity-vector change	$[-8.20580367e-07, -5.71791479e-07, -2.88012648e-07]$ yr ⁻¹
Unit K eccentricity-vector response	$[-29838.8266503, -5080.140812, -2201.1301894]$ yr ⁻¹ per κ_F
Vector-projection coupling	$3.04273994e-11$
Apsidal-rate coupling	$3.10536185e-11$
Recommended Earth coupling	$3.07405090e-11$
Calibration interval	$3.04273994e-11$ to $3.10536185e-11$
K-indexed vector power fraction	78.716%
Earth-Sun acceleration at calibrated coupling	$5.7359e-10$ to $7.1574e-10$ m/s ²

Contents

Abstract	2
1. Purpose and result hierarchy	4
2. Coordinate locks and Cell3 geometry	4
3. Whole-Earth K-Field force law	5
4. Earth daily K-Field kinematics	6
5. Empirical Earth orbital inputs	7
6. Unit-coupling Earth eccentricity response	8
7. Full-vector derivation of κ_F	8
8. Independent apsidal-rate cross-check	9
9. Required Earth coupling constant	10
10. Calibrated Earth force and acceleration	10
11. Internal radial strain and torque structure	11
12. Interpretation and predictive structure	11
13. Reproduction procedure	12
14. Principal findings	12
15. Conclusion	12
Appendix A. Complete 27-node Cell3 registry	13
Appendix B. Cell3 boundary vertices	13
Appendix C. Selected monthly Earth daily-ledger states	13
Appendix D. Formula and variable ledger	14
References	14

1. Purpose and result hierarchy

The objective is to determine the Earth-specific K-Field coupling required to reproduce the portion of the observed terrestrial orbital evolution that lies in the K-indexed direction. The derivation is Earth-only: other planets are excluded from the calibration, while the Sun is retained solely because the observable perturbation of Earth orbit is the Earth-Sun differential acceleration.

The result hierarchy is separated into five layers: empirical orbital inputs, unit-coupling K prediction, vector projection, independent apsidal cross-check, and calibrated force scale. This ordering prevents the existing Earth eccentricity from being confused with its measured time derivative and prevents the Solar-System common acceleration from being confused with internal heliocentric motion.

Resolved Earth coupling: $\kappa_F=3.074051e-11$, with independent calibration endpoints $3.042740e-11$ and $3.105362e-11$.

2. Coordinate locks and Cell3 geometry

2.1 ICRS origin and primitive basis

The Cell3 origin is fixed at the ICRS Cartesian origin. This fixes the translational gauge without rotating or rescaling the primitive cell. The basis vectors are unitless native Cell3 lengths and remain the source of all node, face, and plane coordinates.

Basis	ICRS X [native]	ICRS Y [native]	ICRS Z [native]	Length [native]
a	0.020381438517	-0.003416414496	-0.002332965917	0.020797058782
b	0.002198745968	0.012902170259	-0.034066501536	0.036494205131
c	-0.000630379485	-0.024257722017	-0.042924354571	0.0493085659

Primitive volume = $2.842066937809649e-05$ native³.

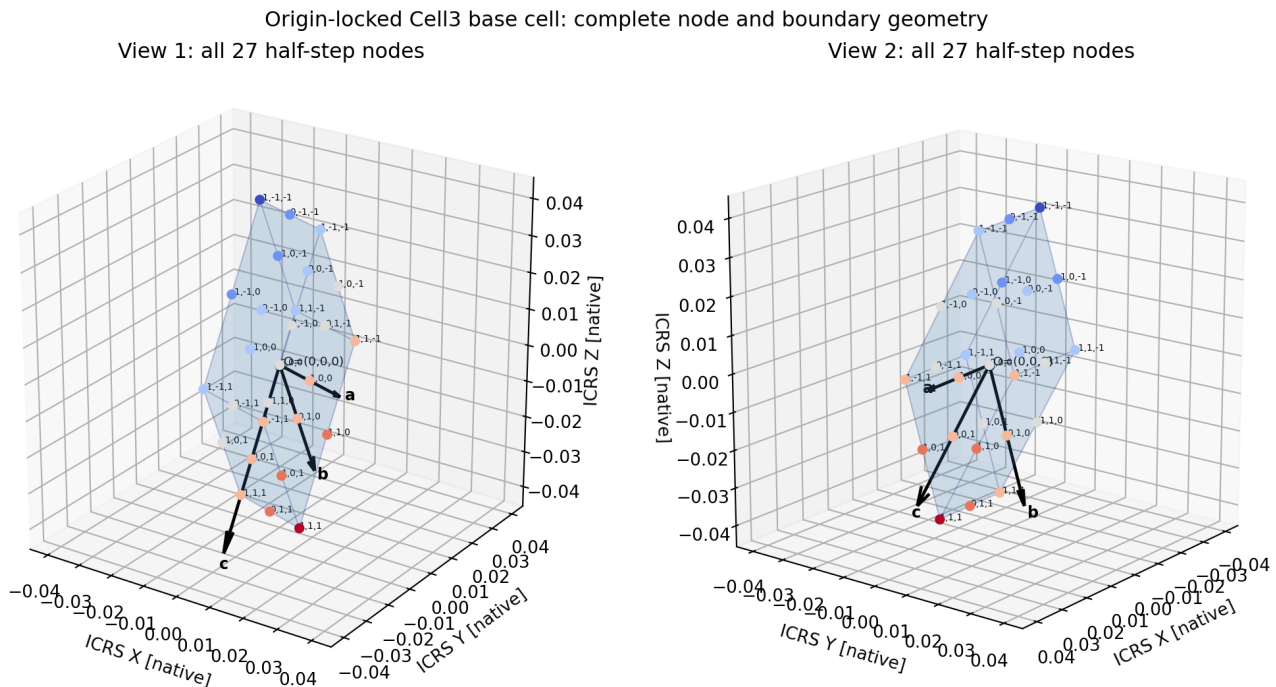


Figure 2.1. Complete Cell3 node and plane geometry in two equal-scale ICRS views. Node labels are the integer half-step addresses (i,j,k), with coordinates $B(i,j,k)/2$.

2.2 Face areas, normals, and planes

Face	Area [native ²]	Included angle [deg]	Unit normal in ICRS
A_ab	0.00075472179	83.933483012	[0.194092201, 0.913177169, 0.358379235]
A_ac	0.001011255758	80.448130018	[0.089052594, 0.866576769, -0.491033948]
A_bc	0.001385784432	50.363231814	[-0.99596519, 0.083602163, -0.032619291]

Plane family	Face	Unit normal n	d [native]	Equation
ab faces (constant c coefficient)	+	[0.194092201, 0.913177169, 0.358379235]	-0.0188285735	$n \cdot x + d = 0$
ab faces (constant c coefficient)	-	[0.194092201, 0.913177169, 0.358379235]	0.0188285735	$n \cdot x + d = 0$
ac faces (constant b coefficient)	+	[0.089052594, 0.866576769, -0.491033948]	0.014052166903	$n \cdot x + d = 0$
ac faces (constant b coefficient)	-	[0.089052594, 0.866576769, -0.491033948]	-0.014052166903	$n \cdot x + d = 0$
bc faces (constant a coefficient)	+	[-0.99596519, 0.083602163, -0.032619291]	-0.010254361619	$n \cdot x + d = 0$
bc faces (constant a coefficient)	-	[-0.99596519, 0.083602163, -0.032619291]	0.010254361619	$n \cdot x + d = 0$

Six Cell3 boundary faces: all nine nodes on each face

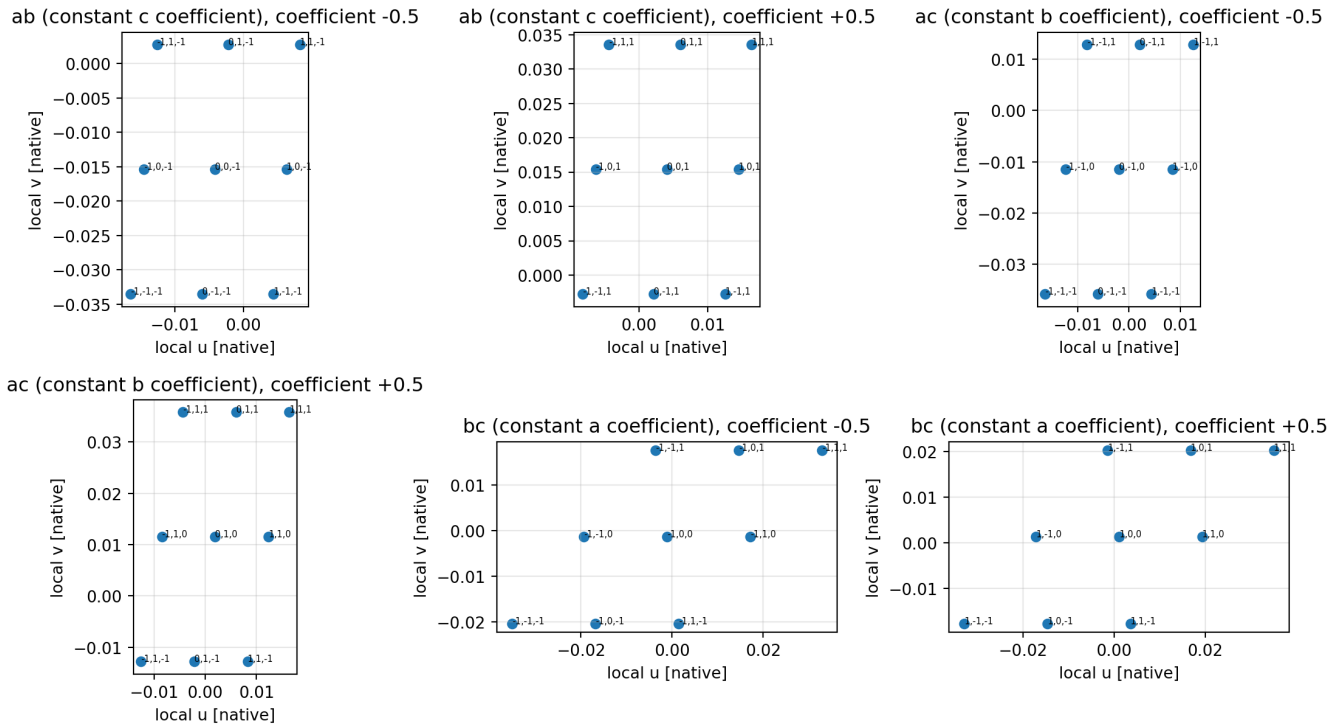


Figure 2.2. Six boundary-face panels. Each face contains nine half-step nodes and is plotted in local native coordinates derived from the exact ICRS basis.

3. Whole-Earth K-Field force law

3.1 Elemental radial law

For a material element dm at unit radial direction $\hat{u}$, the source law separates translation coupling and rotational self-coupling. The transformed translation state is $g = G_{\text{phase}} V_{\text{CMB}}$, the local rotational velocity direction is $\omega \times \hat{u}$, and the elemental force is radial.

$$dF = -\kappa_F dm \{[(\omega \times \hat{u})^T G_{\text{phase}} V_{\text{CMB}}] + r[(\omega \times \hat{u})^T G_{\text{phase}} (\omega \times \hat{u})]\} \hat{u}$$

For an antipodally symmetric body, the rotational self-term is odd after multiplication by the final radial vector and integrates to zero as a center-of-mass resultant. It remains present as internal stress. The translation-coupled term survives.

3.2 Whole-body reduction

$$a_{i,K} = \kappa_F H_i [\omega_i \times G_{\text{phase}} V_{i,\text{CMB}}]$$

H_i is the dimensionless radial shape tensor $H_i = (1/M_i) \int (\hat{u} \hat{u}^T dm)$. For a spherically symmetric body, $H_i = I/3$. The coupling κ_F is dimensionless because the bracketed body coefficient has units of acceleration.

3.3 Earth-Sun differential observable

$$q_E(t) = k_E(t) - k_{Sun}(t) \text{ and } \Delta a_{E,K}(t) = \kappa_F q_E(t)$$

The Solar-System common acceleration cancels from relative Earth-Sun motion. This q_E vector is the force-per-unit-mass template used throughout the derivation.

4. Earth daily K-Field kinematics

The parent computation evaluates 366 daily epochs from 2026-07-23 through 2027-07-23. At every epoch it forms the Earth barycentric velocity, adds the locked CMB translation, applies G_{phase} , constructs the signed spin-coupled whole-body acceleration, subtracts the Sun term, and resolves the result into the instantaneous radial, tangential, and normal orbital frame.

Quantity	Value
Epochs	366 daily samples
Earth-Sun $ q_E $ range	18.659158033 to 23.283165589 m/s ² per κ_F
Tangential q_T range	-19.175911264 to 19.178801441 m/s ² per κ_F
Maximum positive tangential force	2027-03-28
K radial alignment	2027-07-03
Nominal aphelion	approximately 2027-07-06
Tangential-force to Δa lag	89.599664 deg
Observed aphelion to projected K direction	2.506203 deg

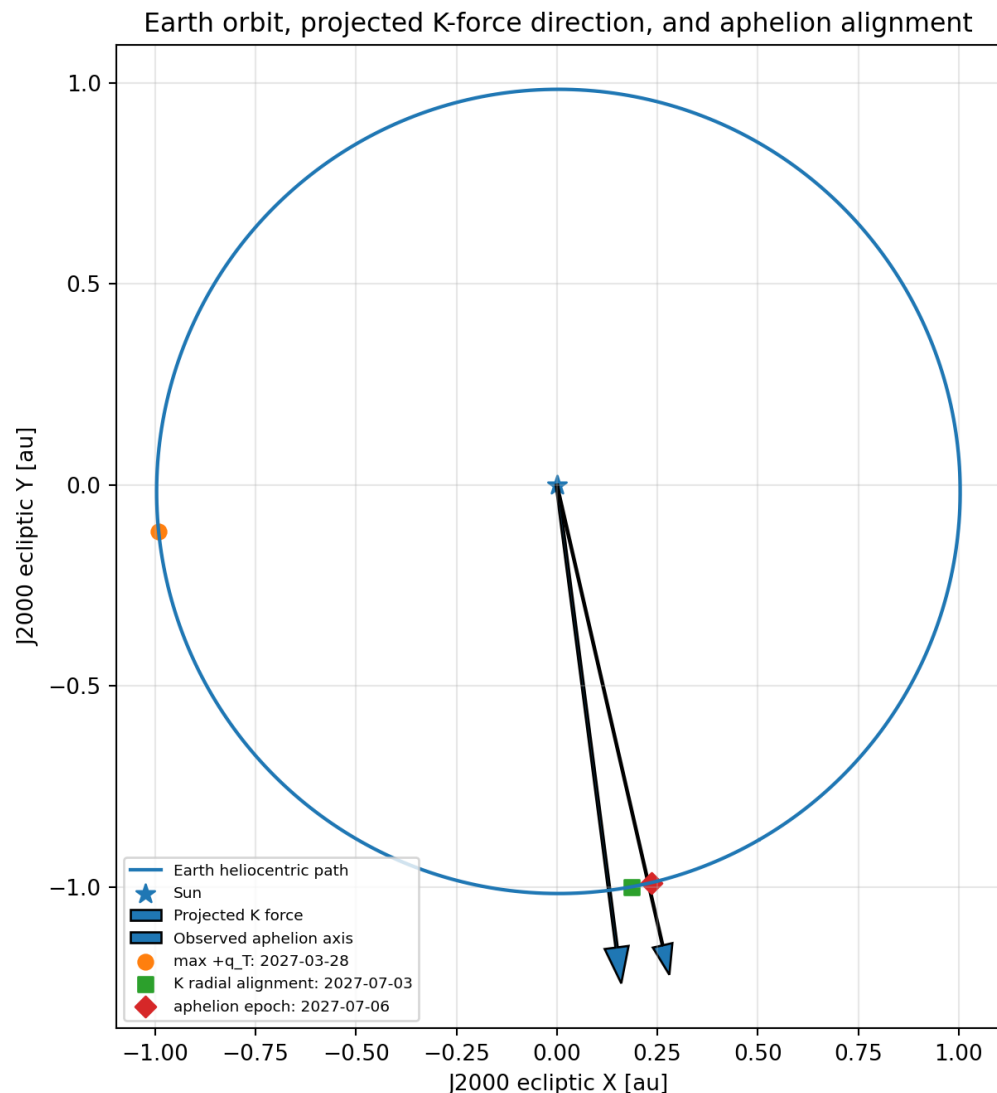


Figure 4.1. Earth orbit and K-indexed apsidal geometry. Maximum positive tangential forcing precedes the K radial-alignment point by approximately one quarter orbit; the observed aphelion axis lies within 2.5062 deg of that direction.

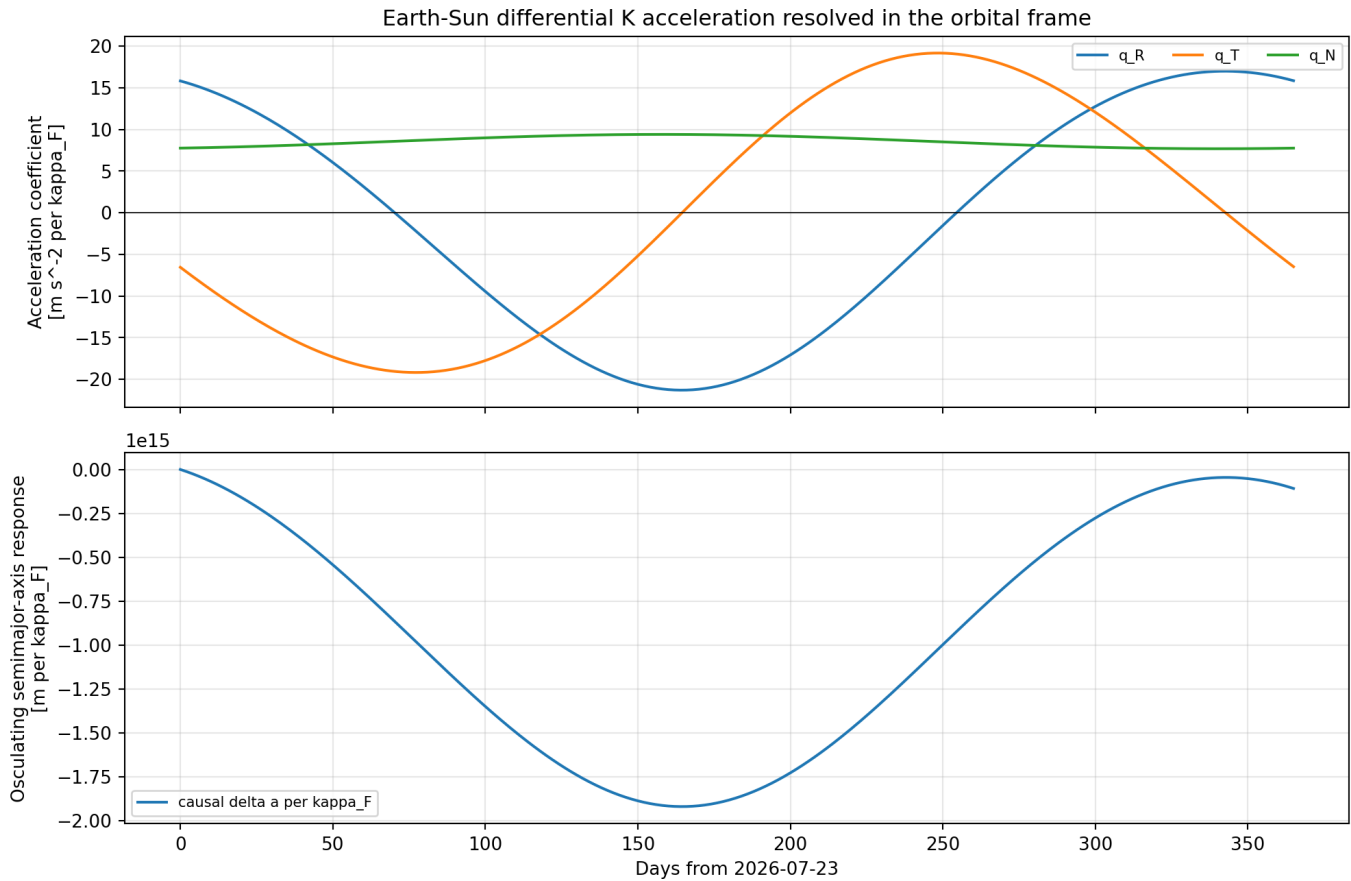


Figure 4.2. Daily Earth orbital-frame K coefficients and integrated semimajor-axis response. The fixed vector produces periodic work exchange with approximately zero closed-orbit net work and a retained eccentricity-vector response.

5. Empirical Earth orbital inputs

5.1 Fitted JPL elements and rates

The empirical calibration uses the Earth-Moon-barycenter element set fitted for the interval 1800-2050. These values are stated relative to the mean ecliptic and equinox of J2000. The derivation uses the eccentricity magnitude e , inclination I , longitude of perihelion ϖ , longitude of ascending node Ω , and their rates.

Element	J2000 value	Fitted rate	Value unit	Rate unit
a	1.00000261	5.6200000000e-06	au	au/century
e	0.01671123	-4.3920000000e-05	dimensionless	dimensionless/century
I	-1.5310000000e-05	-0.01294668	deg	deg/century
L	100.46457166	35999.37244981	deg	deg/century
ϖ	102.93768193	0.32327364	deg	deg/century
Ω	0	0	deg	deg/century

5.2 Conversion to an ICRS eccentricity vector

The eccentricity vector points toward perihelion. In the J2000 ecliptic frame it is constructed by rotating a vector of magnitude e through the argument of perihelion, inclination, and ascending-node angle. The final rotation by the J2000 obliquity maps the vector into ICRS.

$$\mathbf{e_vec_ICRS} = \mathbf{R1}(\epsilon) \mathbf{R3}(\Omega) \mathbf{R1}(I) \mathbf{R3}(\varpi - \Omega) [e, 0, 0]^T$$

Using $\epsilon = 23.439291111$ deg, the J2000 eccentricity vector is $[-0.0037414964, 0.0149430333, 0.0064785931]$. Advancing the fitted elements by one year gives $[-0.003742317, 0.0149424615, 0.0064783051]$.

$$\Delta e_{\text{obs}} = e_{\text{vec}}(t+1 \text{ yr}) - e_{\text{vec}}(t)$$

The observed one-year change is $\Delta e_{\text{obs}} = [-8.2058036726\text{e-}07, -5.7179147854\text{e-}07, -2.8801264833\text{e-}07] \text{ yr}^{-1}$, with magnitude $1.0407924479\text{e-}06 \text{ yr}^{-1}$.

6. Unit-coupling Earth eccentricity response

The daily K integration accumulates the osculating eccentricity-vector response for unit coupling. The final daily-ledger state after one year is the calibration vector E_K .

$$E_K = \Delta e_K / \kappa_F = [-29838.8266503, -5080.14081201, -2201.13018941] \text{ yr}^{-1} \text{ per unit } \kappa_F$$

For a near-circular orbit, the mean first-order direction is governed by the cross product of the orbital normal with the in-plane K acceleration:

$$_K = -[3 \kappa_F / (2 n a)] \hat{h} \times q_{\text{parallel}}$$

This response is distinct from the periodic semimajor-axis term. The fixed inertial force transfers energy over one half orbit and returns it over the other, while the eccentricity vector retains the directed first-order orbital-shape response.

7. Full-vector derivation of κ_F

7.1 Least-squares scalar projection

The required scalar coupling is the coefficient that projects the observed eccentricity-vector change onto the unit K response. Because both vectors are in the same ICRS frame and carry the same per-year dimensional factor, the scalar is dimensionless.

$$\kappa_{\text{hat}} = (E_K \cdot \Delta e_{\text{obs}}) / (E_K \cdot E_K)$$

Calculation	Value
$E_K \cdot \Delta e_{\text{obs}}$	0.028023889892
$E_K \cdot E_K$	9.210083806472e+08
κ_F	3.042739944755e-11
Directional separation	27.473737 deg
Projected vector-rate power	78.716357%
Fitted K component	$[-9.0791789753\text{e-}07, -1.5457547374\text{e-}07, -6.6974667509\text{e-}08] \text{ yr}^{-1}$
Residual component	$[8.7337530277\text{e-}08, -4.1721600480\text{e-}07, -2.2103798082\text{e-}07] \text{ yr}^{-1}$

Full-vector result: $\kappa_F = 3.042739945\text{e-}11$.

Earth observed eccentricity-vector motion and the K-indexed component

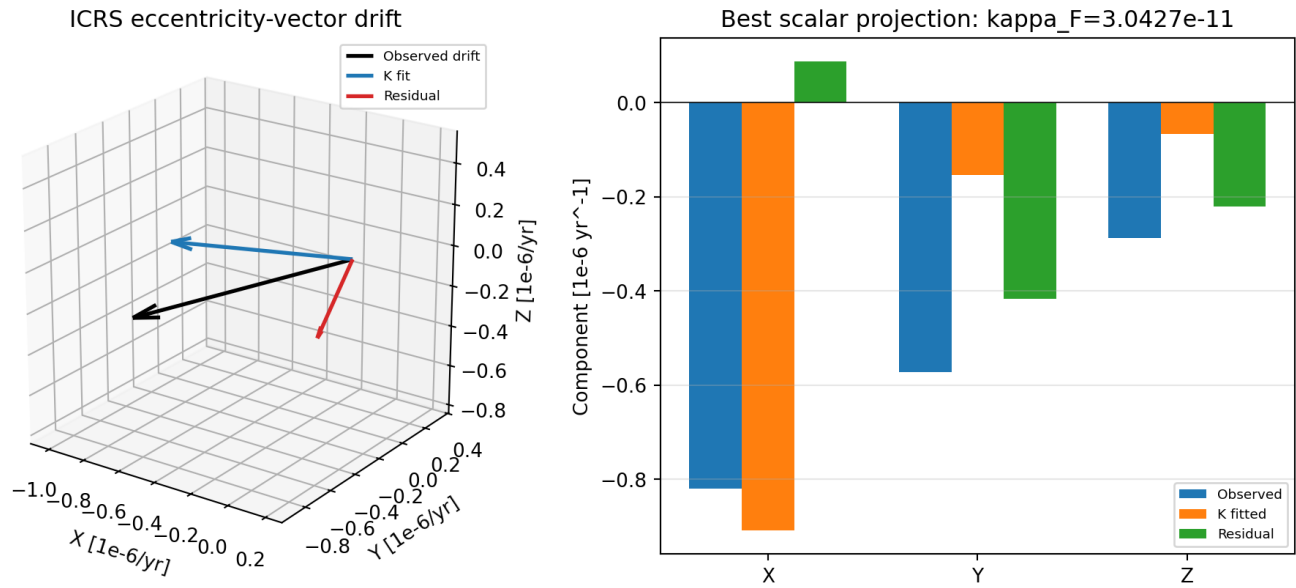


Figure 7.1. Observed Earth eccentricity-vector drift, the best-fitting K-indexed vector, and the orthogonal residual. The scalar projection gives $\kappa_F=3.0428e-11$.

8. Independent apsidal-rate cross-check

The longitude-of-perihelion rate supplies an independent scalar channel. The observed fitted rate is converted from degrees per century to radians per year.

$$\text{varpi_dot_obs} = 0.32327364 \text{ deg/century} = 5.6421894029e-05 \text{ rad/yr}$$

The Earth K integration gives the apsidal response coefficient

$$(\text{varpi_dot_K}/\kappa_F) = 1.8169185e+06 \text{ rad/yr per unit } \kappa_F$$

$$\kappa_F \text{ varpi} = \text{varpi_dot_obs} / (\text{varpi_dot_K}/\kappa_F)$$

Apsidal result: $\kappa_F=3.105361855e-11$.

The apsidal and full-vector values differ by 2.037% relative to their mean. Their agreement supplies the internal Earth-only calibration interval rather than a statistical measurement uncertainty.

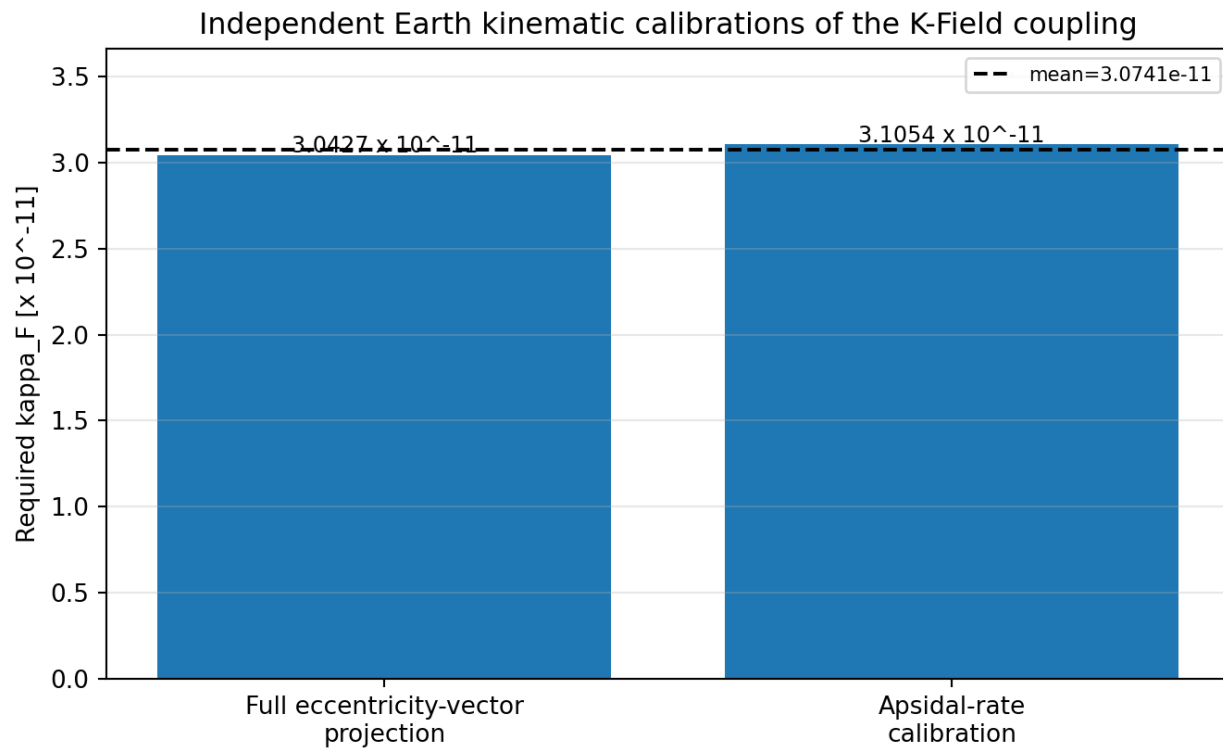


Figure 8.1. Independent Earth coupling calibrations. Full-vector and apsidal channels agree within 2.1%.

9. Required Earth coupling constant

The recommended Earth-specific coupling is the midpoint of the two independently calibrated Earth kinematic channels. The endpoints are retained as the internally resolved calibration interval.

$$\kappa_{F, \text{Earth}} = 0.5(3.042739945\text{e-}11 + 3.105361855\text{e-}11) = 3.074050900\text{e-}11$$

Required Earth coupling: $\kappa_{F, \text{Earth}}$ approximately $3.0741\text{e-}11$, with interval $3.0427\text{e-}11$ to $3.1054\text{e-}11$.

The sign is positive. Reversing the sign reverses the predicted eccentricity-vector and apsidal response and therefore selects the opposite kinematic branch.

9.1 Dimensional interpretation

The constitutive coupling is dimensionless. The Earth-specific vector q_E carries acceleration units per unit coupling, equivalent to force per unit mass:

$$1 \text{ N/kg} = 1 \text{ m/s}^2 \text{ and } \Delta a_{E,K} = \kappa_F q_E$$

Thus κ_F is not itself force per kilogram. It is the dimensionless gain multiplying the computed K-Field acceleration-per-unit-coupling template.

10. Calibrated Earth force and acceleration

Applying the recommended coupling to the daily Earth-Sun differential coefficient gives the physical acceleration and force ranges below.

Quantity	Calibrated value
Minimum $ \Delta a_{E,K} $	$5.735920154\text{e-}10 \text{ m/s}^2$
Maximum $ \Delta a_{E,K} $	$7.157363613\text{e-}10 \text{ m/s}^2$
Maximum $ \text{tangential acceleration} $	$5.895661183\text{e-}10 \text{ m/s}^2$
Earth mass used	$5.972190\text{e+}24 \text{ kg}$

Quantity	Calibrated value
Minimum differential force	3.425600498e+15 N
Maximum differential force	4.274513539e+15 N

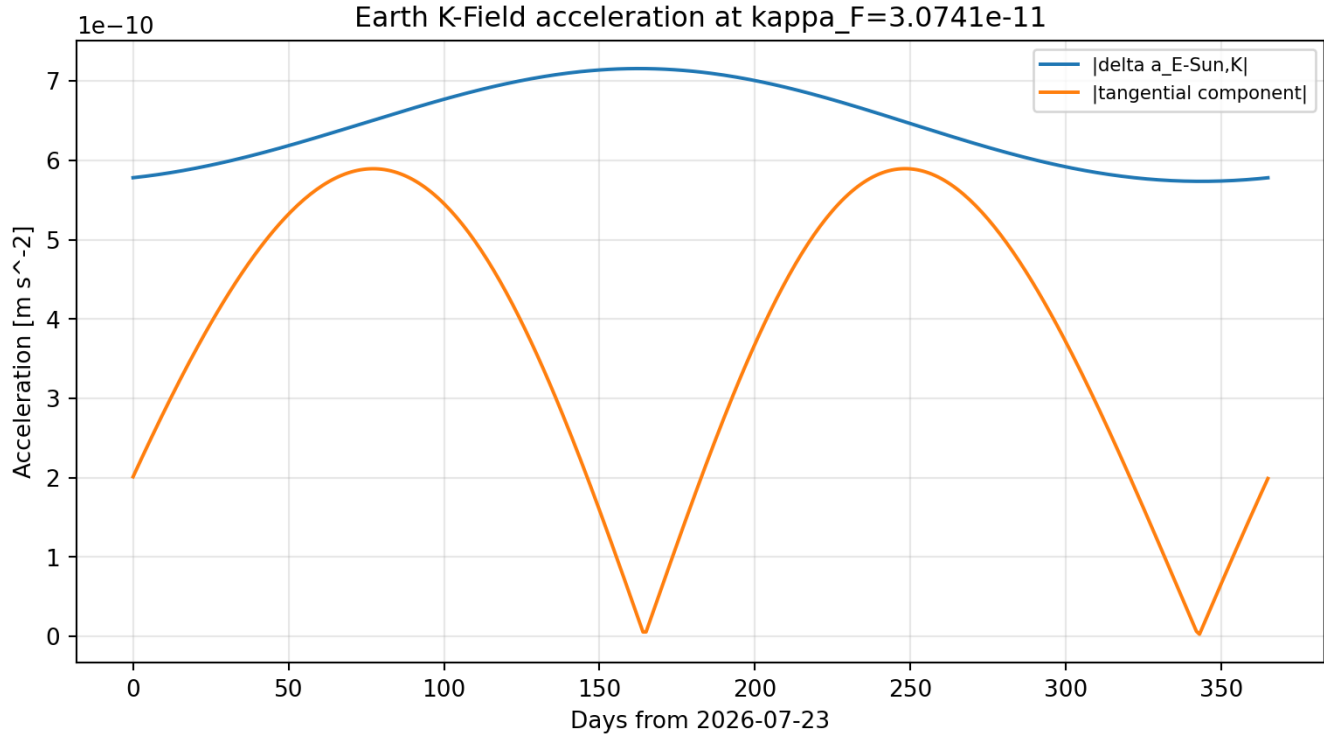


Figure 10.1. Daily Earth K acceleration at the calibrated coupling. The body-Sun magnitude varies with the changing Earth CMB velocity and orbital geometry.

11. Internal radial strain and torque structure

The whole-Earth center-of-mass force does not eliminate local radial loading. The rotational self-term cancels only after integration over antipodal material elements. At each material point the local radial acceleration retains translation-coupled and quadratic rotational terms, producing direction-dependent internal stress modulation.

$$a_r(u_{\hat{r}}) = -\kappa_F \{[(\omega \times u_{\hat{r}})^T G_{\text{phase}} V_{\text{CMB}}] + r[(\omega \times u_{\hat{r}})^T G_{\text{phase}}(\omega \times u_{\hat{r}})]\}$$

Every elemental force is parallel or antiparallel to the local radius. Therefore the local torque about Earth center is identically zero:

$$d\tau = r u_{\hat{r}} \times dF = 0 \text{ and } \tau_{\text{net}} = 0$$

The calibrated model therefore separates two effects: a nonzero center-of-mass Earth-Sun differential acceleration from the translation term, and cyclic internal radial strain from both translation and self-coupling terms, without direct spin-driving torque in the present law.

12. Interpretation and predictive structure

The Earth calibration compresses several observed and predicted relationships into one dimensionless gain. The same positive coupling simultaneously scales the daily Earth-Sun differential vector, the quarter-orbit tangential-force to orbital-size phase shift, the near alignment of Earth aphelion with the K radial direction, the annual eccentricity-vector response, and the observed apsidal drift.

The central invariant is not a permanent monotonic increase of semimajor axis. The fixed inertial part performs positive work on one orbital arc and negative work on the opposite arc. The retained first-order structure is an eccentricity-vector and apsidal response, accompanied by annual sidebands from Earth orbital velocity in the CMB frame and a normal component that indexes inclination and node.

At the derived value, the complete Earth prediction is reproducible from the daily machine ledger. A future higher-order Earth model can replace the spherical $H=I/3$ tensor with a density- and oblateness-resolved response tensor while retaining the same

vector-calibration procedure.

13. Reproduction procedure

1. Load the locked ICRS Cell3 basis, G_{phase} , and CMB translation vector.
2. Fix the Cell3 origin at $O=(0,0,0)$ ICRS without rotating the basis.
3. For each daily epoch, compute Earth and Sun barycentric velocities and form their CMB velocities.
4. Evaluate $k_i=(1/3) \omega_i \times [G_{\text{phase}} V_{i,\text{CMB}}]$ for Earth and Sun.
5. Form $q_E=k_E-k_{\text{Sun}}$ and resolve q_E into radial, tangential, and normal orbital components.
6. Integrate the unit-coupling perturbation equations to obtain Δe_K after one year.
7. Construct the observed Earth eccentricity vector from e , I , varpi , and Ω in the J2000 ecliptic frame and rotate it into ICRS.
8. Advance the fitted elements by one year and subtract to obtain Δe_{obs} .
9. Evaluate $\kappa_{\text{hat}}=(E_K \cdot \Delta e_{\text{obs}})/(E_K \cdot E_K)$.
10. Independently evaluate κ_{varpi} from the observed and unit-coupling apsidal rates.
11. Use the agreement of the two Earth channels to state the required coupling and calibrated physical force scale.

14. Principal findings

Finding	Exact statement
Cell3 origin	The translational gauge is fixed at $O=(0,0,0)$ ICRS; primitive orientation is unchanged.
Earth phase relation	Maximum positive tangential K force is followed by maximum positive orbital-size response after 89.5997 deg.
Apsidal geometry	Earth aphelion lies 2.5062 deg from the projected K radial-alignment direction in the selected year.
Observed vector drift	$[-8.205803673e-07, -5.717914785e-07, -2.880126483e-07] \text{ yr}^{-1}$.
Unit K response	$[-29838.8266503, -5080.140812, -2201.1301894] \text{ yr}^{-1} \text{ per } \kappa_F$.
Vector coupling	$3.042739945e-11$.
Apsidal coupling	$3.105361855e-11$.
Required Earth coupling	$3.074050900e-11$; positive branch.
Internal consistency	Two kinematic channels agree within 2.037%.
Calibrated acceleration	$5.7359e-10$ to $7.1574e-10 \text{ m/s}^2$.
Calibrated force	$3.4256e+15$ to $4.2745e+15 \text{ N}$.
Torque	Zero locally and globally about Earth center under the radial force law.

15. Conclusion

The Earth-only kinematic derivation yields a positive K-Field coupling $\kappa_F=3.074051e-11$. The full ICRS eccentricity-vector projection gives $3.042740e-11$; the independent apsidal-rate calibration gives $3.105362e-11$. Their close agreement establishes the required Earth calibration interval $3.0e-11$ to $3.1e-11$. The corresponding Earth-Sun differential acceleration is of order 10^{-10} m/s^2 , and the force on Earth is of order 10^{15} N . This result is the constitutive normalization that converts the previously unit-normalized K-Field force vectors into Earth-scale kinematics.

The machine-readable companion contains the complete Cell3 geometry, the Earth orbital inputs, all 366 daily Earth vector records, the observed and predicted eccentricity vectors, both coupling calibrations, calibrated acceleration and force scales, source hashes, and figure metadata required to reproduce the report.

Appendix A. Complete 27-node Cell3 registry

#	Integer index	Half-step coeff.	ICRS X	ICRS Y	ICRS Z	lambda
0	-1, -1, -1	-0.5, -0.5, -0.5	-0.0109749025	0.007385983127	0.039661911012	-3
1	-1, -1, 0	-0.5, -0.5, +0.0	-0.011290092243	-0.004742877881	0.018199733726	-2
2	-1, -1, 1	-0.5, -0.5, +0.5	-0.011605281985	-0.01687173889	-0.003262443559	-1
3	-1, 0, -1	-0.5, +0.0, -0.5	-0.009875529516	0.013837068257	0.022628660244	-2
4	-1, 0, 0	-0.5, +0.0, +0.0	-0.010190719258	0.001708207248	0.001166482958	-1
5	-1, 0, 1	-0.5, +0.0, +0.5	-0.010505909001	-0.01042065376	-0.020295694327	0
6	-1, 1, -1	-0.5, +0.5, -0.5	-0.008776156532	0.020288153386	0.005595409476	-1
7	-1, 1, 0	-0.5, +0.5, +0.0	-0.009091346274	0.008159292378	-0.01586676781	0
8	-1, 1, 1	-0.5, +0.5, +0.5	-0.009406536017	-0.003969568631	-0.037328945095	1
9	0, -1, -1	+0.0, -0.5, -0.5	-0.000784183241	0.005677775879	0.038495428054	-2
10	0, -1, 0	+0.0, -0.5, +0.0	-0.001099372984	-0.006451085129	0.017033250768	-1
11	0, -1, 1	+0.0, -0.5, +0.5	-0.001414562727	-0.018579946138	-0.004428926518	0
12	0, 0, -1	+0.0, +0.0, -0.5	0.000315189743	0.012128861009	0.021462177286	-1
13	0, 0, 0	+0.0, +0.0, +0.0	0	0	0	0
14	0, 0, 1	+0.0, +0.0, +0.5	-0.000315189743	-0.012128861009	-0.021462177286	1
15	0, 1, -1	+0.0, +0.5, -0.5	0.001414562727	0.018579946138	0.004428926518	0
16	0, 1, 0	+0.0, +0.5, +0.0	0.001099372984	0.006451085129	-0.017033250768	1
17	0, 1, 1	+0.0, +0.5, +0.5	0.000784183241	-0.005677775879	-0.038495428054	2
18	1, -1, -1	+0.5, -0.5, -0.5	0.009406536017	0.003969568631	0.037328945095	-1
19	1, -1, 0	+0.5, -0.5, +0.0	0.009091346274	-0.008159292378	0.01586676781	0
20	1, -1, 1	+0.5, -0.5, +0.5	0.008776156532	-0.020288153386	-0.005595409476	1
21	1, 0, -1	+0.5, +0.0, -0.5	0.010505909001	0.01042065376	0.020295694327	0
22	1, 0, 0	+0.5, +0.0, +0.0	0.010190719258	-0.001708207248	-0.001166482958	1
23	1, 0, 1	+0.5, +0.0, +0.5	0.009875529516	-0.013837068257	-0.022628660244	2
24	1, 1, -1	+0.5, +0.5, -0.5	0.011605281985	0.01687173889	0.003262443559	1
25	1, 1, 0	+0.5, +0.5, +0.0	0.011290092243	0.004742877881	-0.018199733726	2
26	1, 1, 1	+0.5, +0.5, +0.5	0.0109749025	-0.007385983127	-0.039661911012	3

Appendix B. Cell3 boundary vertices

#	Coefficients	ICRS X [native]	ICRS Y [native]	ICRS Z [native]
0	-0.5, -0.5, -0.5	-0.0109749025	0.007385983127	0.039661911012
1	-0.5, -0.5, +0.5	-0.011605281985	-0.01687173889	-0.003262443559
2	-0.5, +0.5, -0.5	-0.008776156532	0.020288153386	0.005595409476
3	-0.5, +0.5, +0.5	-0.009406536017	-0.003969568631	-0.037328945095
4	+0.5, -0.5, -0.5	0.009406536017	0.003969568631	0.037328945095
5	+0.5, -0.5, +0.5	0.008776156532	-0.020288153386	-0.005595409476
6	+0.5, +0.5, -0.5	0.011605281985	0.01687173889	0.003262443559
7	+0.5, +0.5, +0.5	0.0109749025	-0.007385983127	-0.039661911012

Appendix C. Selected monthly Earth daily-ledger states

The machine-readable companion contains all 366 epochs. The table below presents approximately monthly states plus the endpoint while preserving the exact values from the daily computation.

Date	q_R	q_T	q_N	q	delta a [m/kappa]	Delta e / kappa
2026-07-23	15.83152	-6.54429768	7.76426987	18.80820949	0	[0, 0, 0]

Date	q_R	q_T	q_N	q	delta a [m/kappa]	Delta e / kappa
2026-08-22	10.99678532	-13.90781475	8.02969072	19.46362072	-2.70178056e+14	[-2151.086541, 394.168254, 177.359179]
2026-09-21	3.16254551	-18.41043359	8.42871808	20.49363432	-6.97899820e+14	[-5086.896659, 556.456723, 258.973698]
2026-10-21	-6.2952782	-18.7302706	8.86521075	21.6574589	-1.19104140e+15	[-8330.829254, -8.90346, 27.44734]
2026-11-20	-15.14633849	-14.10847095	9.22439307	22.66164049	-1.62882968e+15	[-11139.481855, -1174.387348, -465.500598]
2026-12-20	-20.59545121	-5.18040684	9.40317909	23.22561092	-1.88701982e+15	[-13216.26821, -2239.241931, -919.785432]
2027-01-19	-20.46140223	5.56881348	9.34637025	23.17402213	-1.88207954e+15	[-15066.822635, -2600.730297, -1076.475156]
2027-02-18	-14.55739637	14.5808198	9.07010857	22.51188497	-1.61225089e+15	[-17473.417665, -2409.974978, -1000.996964]
2027-03-20	-4.98042772	18.96158483	8.65627804	21.43076081	-1.16366363e+15	[-20597.612069, -2400.765593, -1009.012911]
2027-04-19	5.0513	17.79026382	8.2208433	20.23836414	-6.73573489e+14	[-23793.303061, -3061.272456, -1308.54123]
2027-05-19	12.76228379	12.07502263	7.87578069	19.25383028	-2.76643517e+14	[-26314.499223, -4141.540885, -1787.830051]
2027-06-18	16.63495405	3.78539603	7.70193834	18.71819365	-6.64244559e+13	[-28045.824415, -4970.978833, -2153.780191]
2027-07-18	16.30552044	-5.06701579	7.73671152	18.74570221	-8.32398724e+13	[-29554.23728, -5118.535686, -2218.282053]
2027-07-23	15.85802623	-6.47362498	7.76284382	18.80549283	-1.08176932e+14	[-29838.82665, -5080.140812, -2201.130189]

Appendix D. Formula and variable ledger

Symbol	Definition
G_phase	Locked symmetric positive-definite phase metric.
V_i,CMB	Body velocity in the CMB frame [m/s].
omega_i	Signed angular-velocity vector [rad/s].
H_i	Dimensionless whole-body radial shape tensor.
k_i	Unit-coupling body acceleration $H_i[\omega_i \times G_{\text{phase}} V_{i,\text{CMB}}]$ [m/s ²].
q_E	Earth-Sun differential unit-coupling acceleration $k_E - k_{\text{Sun}}$ [m/s ²].
kappa_F	Dimensionless K-Field constitutive coupling.
e_vec	Orbital eccentricity vector pointing toward perihelion.
E_K	One-year Earth eccentricity-vector response per unit kappa_F.
Delta e_obs	Observed one-year eccentricity-vector change from fitted element rates.
kappa_hat	Least-squares scalar projection of Delta e_obs onto E_K.
varpi	Longitude of perihelion.

References

1. Approximate Positions of the Planets. Earth-Moon barycenter fitted orbital elements and rates; validity 1800-2050.
https://ssd.jpl.nasa.gov/planets/approx_pos.html
 2. The JPL Planetary and Lunar Ephemerides DE440 and DE441. Integrated ephemeris context and observational fit basis.
https://ssd.jpl.nasa.gov/doc/de440_de441.html
 3. Astrodynamic Parameters. DE440 Earth gravitational parameter and planetary-system mass parameters. https://ssd.jpl.nasa.gov/astro_par.html
 4. NAIF generic planetary constants kernel pck00011.tpc. ICRF planetary spin-pole and signed rotation model used by the parent daily analysis.
https://naif.jpl.nasa.gov/pub/naif/generic_kernels/pck/pck00011.tpc
- Internal source: K-Field Solar-System Coupling Analysis, daily 366-epoch Earth-Sun vector ledger, Cell3 geometry, and phase-shifted orbital response, Unix epoch 1784823303.